

Satellite Inference

A 10 kW continuous-compute node for mission-assured processing in low Earth orbit

Compute where space data begins.



Notional concept render. This is not flight CAD and does not establish interfaces, dimensions or launch compatibility.

The first owned flight target is a 10 kW continuous compute system in LEO. A 1 kW unit remains a ground engineering tile, not a separate orbital product. The current development-capital target is \$7 million.

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Satellite Inference™ is currently operated by RFID INC, a Delaware corporation.

Version: Public Whitepaper v0.2 | Rev C

Date: 2 September 2026

Evidence cut-off: 2 September 2026

EVIDENCE DISCIPLINE

How to read this document

This version replaces the earlier small hosted-flight ladder. It is intentionally ambitious, but it separates requirements, calculations, assumptions and hypotheses so that ambition is not mistaken for completion.

Label	Meaning	What closes it
REQUIREMENT	A program choice that the design must satisfy.	Formal allocation and change control.
CALCULATED	A result of the public Rev C screening equations.	Independent calculation and later correlated analysis.
PUBLIC FACT	A claim supported by a linked public source.	Source review and evidence cut-off.
WORKING ASSUMPTION	A provisional value used to size the concept.	Supplier ROM, analysis or test.
BUSINESS HYPOTHESIS	A claim about buyer need, price or utilization.	Paid evaluation, reservation or contract.

Fixed program decisions

- 10 kW means continuous electrical input at the compute-payload DC interface during nominal operation.
- The total spacecraft demand is provisionally 12-15 kW, including platform, communications and thermal transport.
- The baseline is 500-600 km LEO. GEO is an educational comparison and possible future customer-specific architecture, not a mission stage.
- Initial buyers are defense and sovereign organizations, EO operators, maritime-domain users and disaster-response programs.
- Generic cloud inference is later. The company must first prove positional and mission value, not sell undifferentiated GPU-hours.

PROGRAM THESIS

1. Executive summary

10 kW

CONTINUOUS PAYLOAD INPUT

550 km

REFERENCE LEO

\$7M

CURRENT CAPITAL TARGET

Satellite Inference proposes a shared orbital edge-processing node that receives data created in space, executes customer-approved inference and data-reduction workloads, then sends compact decision products to Earth or to another spacecraft. The purpose is to shorten time-to-decision, preserve scarce communications capacity and enable cross-cueing where ground connectivity is limited or contested.

The first flight target is intentionally large enough to matter: 10 kW continuous at the compute-payload interface. That requirement places the concept near the edge of publicly advertised commercial platform capabilities. Neither the spacecraft bus, the heat-rejection system nor a launch arrangement is yet selected. The program therefore begins with a 1 kW ground tile and a 10 kW integrated ground breadboard, followed by supplier-backed SRR and PDR evidence.

FINANCING
LOGIC

The \$7 million round is designed to buy engineering evidence, customer evidence and program definition. It is not presented as sufficient to manufacture and launch the complete 10 kW spacecraft.

What success looks like

- A repeatable 1 kW compute, networking, power and cooling tile with measured performance and fault recovery.
- A 10 kW ground system that runs representative customer workloads continuously and closes an auditable power and thermal balance.
- Supplier-backed ROMs and interfaces for bus, arrays, battery, heat transport, radiators, communications, integration, test and launch.
- At least two paid evaluations and a conditional anchor reservation tied to workload, data path, performance and price.
- SRR and PDR packages that support a credible next financing and not a speculative mass-production claim.

MARKET WEDGE

2. The initial customer problem

The product is not a sensor and not a generic cloud region. A sensor is the instrument that creates the original data, such as an optical camera, thermal or hyperspectral imager, synthetic-aperture radar or RF receiver. Inference is the execution of an already-trained model against that data. Downlink is transmission to Earth; crosslink is transmission between spacecraft.

Priority workload	Data source	Decision product	Likely economic buyer
Defense and maritime intelligence	Optical, SAR, RF, AIS	Vessel track, anomaly or change alert	Defense ministry, coast guard, intelligence customer or prime
EO operator acceleration	Raw or near-raw optical, hyperspectral and SAR	Quality screen, selected chip, compression or change map	EO constellation operator or analytics provider
Disaster detection and cross-cueing	Thermal, multispectral, SAR and weather	Fire, flood or storm alert plus follow-up task	Emergency agency, utility, insurer or government program
Generic AI inference	Data uploaded from Earth	Tokens, embeddings or general model output	Deferred until orbital economics and fleet utilization improve

Why the buyer pays

- Shorter time from collection to a usable decision.
- Priority and assured processing during operationally important periods.
- Reduction of raw data that must be downlinked.
- Autonomous detection and cross-tasking of another spacecraft.
- A security, sovereignty or remote-area outcome that ordinary cloud compute alone cannot provide.

GAO reported \$76.8 million of JCO purchases through the Global Data Marketplace from January 2023 through September 2025. That is adjacent buyer-budget evidence, not proof that those buyers will purchase independent orbital compute. BlackSky's nearly \$30 million one-year assured contract is evidence that validated mission outcomes can command substantial commitments, not a price benchmark for Satellite Inference.

PRINCIPAL

COMMERCIAL

UNKNOWN

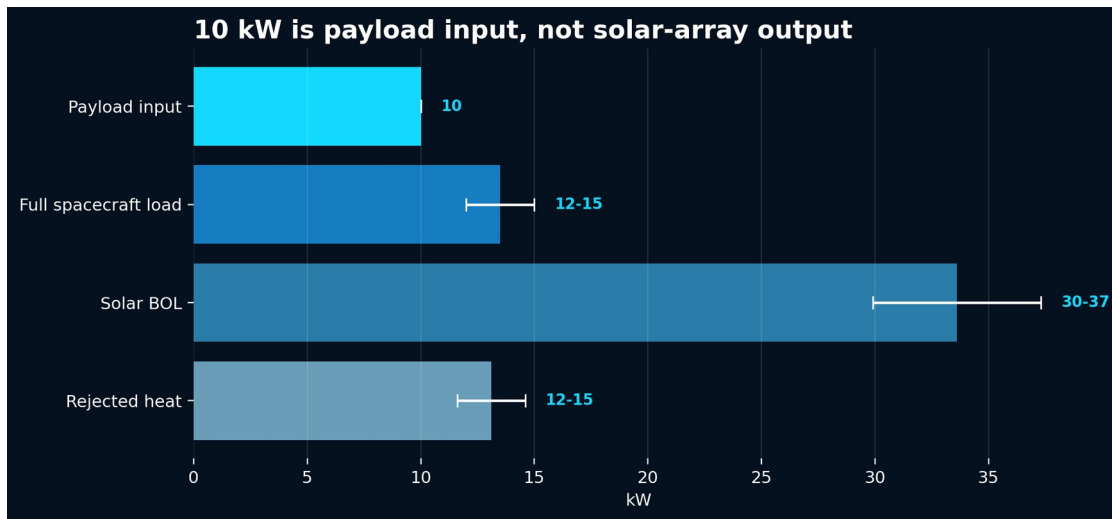
No public procurement found establishes demand for continuous third-party 10 kW LEO capacity. The power level is an engineering platform target until customer datasets, sustained load profiles and paid reservations prove otherwise.

SYSTEM BOUNDARY

3. Product definition and data flow

10.0 kW	12-15 kW	7-9 kW
COMPUTE DC INPUT	FULL SPACECRAFT LOAD	ILLUSTRATIVE ACCELERATOR SHARE

The 10 kW requirement includes accelerators, host CPUs, memory, storage and compute-side conversion losses. It does not claim 10 kW of GPU silicon. Depending on hardware selection, accelerator consumption could be approximately 7-9 kW. That split is a working assumption and will be replaced by a workload-specific bill of materials.



Rev C power definitions. Bars show requirements or current planning ranges, not qualified flight values.

Operational chain

Step	Function	Example output
1. Sensor	Creates raw observation data on a partner spacecraft or hosted instrument.	Image, radar data, RF sample or telemetry stream
2. Data ingress	Moves selected data to the node through a same-platform interface or crosslink.	Prioritized data packet with provenance
3. Orbital processing	Runs approved models, fusion, compression and workflow logic.	Detection, classification, track or selected chip
4. Delivery	Downlinks a compact result or cross-tasks another asset.	Alert, map, message or follow-up task

A free-flying node without contracted sensor ingress or a validated optical or RF data path is idle infrastructure. Data rights, interfaces and network availability are first-order requirements, not follow-on software details.

BUILD EVIDENCE BEFORE FLIGHT

4. Development architecture

Stage	Purpose	Exit evidence	Capital role
1 kW Ground Engineering Tile	Reusable compute, networking, power and cooling building block	Continuous workload, fault recovery, measured electrical and thermal performance	Seed
10 kW Integrated Ground Breadboard	Ten-tile system and full-load facility test	Sustained 10 kW workload, heat transport, orchestration, security and customer benchmarks	Seed
10 kW Orbital Edge Node	First owned LEO system	Commissioning, staged power ramp, sensor-to-result service and disposal readiness	Flight-development and program financing
100 kW Orbital Module	Replicated high-power architecture	Only after first-node operations and contracted expansion	Later growth capital
Megawatt Orbital Network	Fleet-level aggregate capacity	Standardized manufacturing and multi-customer network economics	Infrastructure-scale capital

ARCHITECTURE

PRINCIPLE

The 1 kW tile is not a detour. It is the replaceable unit through which the company proves compute, power, thermal, networking and software behavior before integrating a multi-tonne spacecraft.

Owned versus purchased

Build and control	Buy or co-develop
Workload runtime, scheduling and telemetry	Spacecraft bus and primary structure
Fault containment, recovery and observability	Deployable solar arrays and battery cells or packs
Compute-tile electrical and data interface	Heat-transport hardware and radiator mechanisms
Customer integration, benchmarking and acceptance	ADCS, propulsion, TT&C, optical crosslink and launch integration
Configuration, evidence ledger and mission data plane	Environmental test services and regulatory counsel

LEO BASELINE

5. Reference orbit and alternatives

Orbit case	One-way light time	Power and battery effect	Program status
550 km circular LEO	About 1.8 ms overhead at zenith	Beta-zero screen: 35.6 min eclipse in a 95.65 min orbit	Baseline screening case
Dawn-dusk SSO	Similar LEO latency	Can reduce eclipse and array storage burden, but needs annual beta, launch-plane and thermal analysis	Conditional trade, not frozen
GEO at 35,786 km altitude	About 119 ms one way to zenith before network and processing	Long eclipse seasons but nearly fixed geometry and long link distance	Comparison only
Sun-Earth L1, about 1.5 million km	About 5 seconds one way before routing	Deep-space communications and operations regime	Outside current program

LEO is the correct starting point because the target data originates in or near LEO, latency is low, launch and operations paths are familiar, and passive deorbit is materially more tractable than from GEO. GEO may become relevant for customer-specific persistent coverage, relay or space-domain-awareness workloads, but it is not a shortcut to the first mission.

Dawn-dusk caution

At 550 km, the cylindrical eclipse screen reaches zero only while absolute beta angle is approximately 67.0 degrees or higher. A dawn-dusk local-time label alone does not prove year-round zero eclipse. A propagated annual beta history, attitude geometry, launch date and maintenance plan are required.

TRANSPARENT FIRST-ORDER MODEL

6. Rev C power, energy and thermal screen

Input or factor	Rev C value	Status
Compute-payload electrical input	10.0 kW continuous	REQUIREMENT
Total spacecraft load	12-15 kW; nominal 13.5 kW	WORKING ASSUMPTION
Solar design margin / path efficiency / EOL retention	1.10 / 0.88 / 0.80	WORKING ASSUMPTION
Active-PV BOL density / gross packing	300 W/m ² / 0.80	WORKING ASSUMPTION
Battery DoD / discharge efficiency / EOL retention / reserve	0.50 / 0.92 / 0.80 / 1.30	WORKING ASSUMPTION
Heat fraction / net radiator rejection	0.97 / 350 W/m ²	WORKING ASSUMPTION

Beta-zero 550 km results

Metric	12 kW case	13.5 kW nominal	15 kW case
Required solar BOL	29.87 kW	33.61 kW	37.34 kW
Active-PV equivalent	99.57 m ²	112.02 m ²	124.46 m ²
Gross solar planform	124.46 m ²	140.02 m ²	155.58 m ²
Battery nameplate	25.16 kWh	28.31 kWh	31.45 kWh
Heat to reject	11.64 kW	13.10 kW	14.55 kW
Equivalent radiator area	33.26 m ²	37.41 m ²	41.57 m ²

The public planning envelope rounds these outputs to 30-38 kW BOL solar generation, 125-156 m² gross solar planform, 25-32 kWh battery capacity and 33-42 m² of equivalent radiator area. A separate 35-60 m² physical radiator planform and 3-7 tonne wet mass remain working assumptions because geometry, structure, plumbing, shielding, propellant and supplier allocations are not yet closed.

Battery cycle-life sensitivity

Allowed depth of discharge	Nominal battery nameplate	Interpretation
50%	28.3 kWh	Current Rev C screen; potentially aggressive for long cycle life
40%	35.4 kWh	Moderate sensitivity
30%	47.2 kWh	Shallow cycling sensitivity for roughly 27,000 five-year eclipses

The applicable depth of discharge cannot be selected from a generic table. Battery chemistry, cell temperature, discharge rate, redundancy and life testing must close the design.

CRITICAL**CONSTRAINT**

Vacuum is not a heat sink by conduction. Nearly every watt used by compute becomes heat that must reach radiating surfaces. Thermal architecture, not GPU availability, is likely the dominant first-node design constraint.

THE NETWORK IS PART OF THE SPACECRAFT

7. Communications, compute and security

Communications

- S-band or equivalent low-rate TT&C for command, health and recovery.
- A high-rate downlink sized to compact products, software loads and selected raw-data contingencies.
- Optical or high-rate RF crosslink for shared-node ingress if the sensor is on another spacecraft.
- Store-and-forward with explicit data aging, priority and deletion policies.
- A ground segment that can route workloads, keys, telemetry and customer results under role-based controls.

Compute architecture

- Tile-level isolation so one workload, accelerator or power converter cannot collapse the node.
- ECC, memory scrubbing, watchdogs, current limiting, brownout recovery and cold-spare policy.
- Signed workloads, measured boot, encrypted storage and hardware-rooted identity where available.
- Checkpointing and graceful power throttling during thermal, communications or attitude constraints.
- Customer-visible service metrics: accepted outputs, p95 latency, false-alarm control, availability and data-reduction ratio.

The final crosslink vendor, frequency plan, antenna, modem, encryption boundary and security accreditation are unresolved. No external-node business case should be approved until end-to-end data ingress is demonstrated and costed.

A STAGED ON-ORBIT COMMISSIONING PLAN

8. First-node mission and success criteria

Phase	Power state	Required evidence
Safe commissioning	Bus loads only	Positive control, thermal telemetry, array deployment, battery and TT&C health
Tile activation	1 kW compute	One tile runs representative workload and recovers from injected faults
Partial cluster	3-5 kW compute	Multi-tile scheduling, heat transport and data movement remain stable
Full nominal	10 kW continuous compute	Sustained operation through the design orbit and all nominal thermal cases
Service demonstration	Customer-defined duty cycle	Sensor-to-result flow, p95 SLA, accuracy and customer acceptance
End-of-life readiness	Reduced and safe states	Disposal and passivation capability verified before degradation becomes limiting

Mission-level pass criteria

- 10 kW continuous payload input is demonstrated for at least seven consecutive days without violating approved electrical or thermal limits.
- The node completes at least one real customer-defined workload with accepted quality and latency.
- Fault injection proves containment, restart and telemetry at tile, network and power-domain levels.
- The data path proves contractual rights, provenance, encryption and deletion behavior.
- Disposal and passivation remain available after the mission's credible worst-case failure set.

CAPITAL FOLLOWS EVIDENCE

9. Development gates

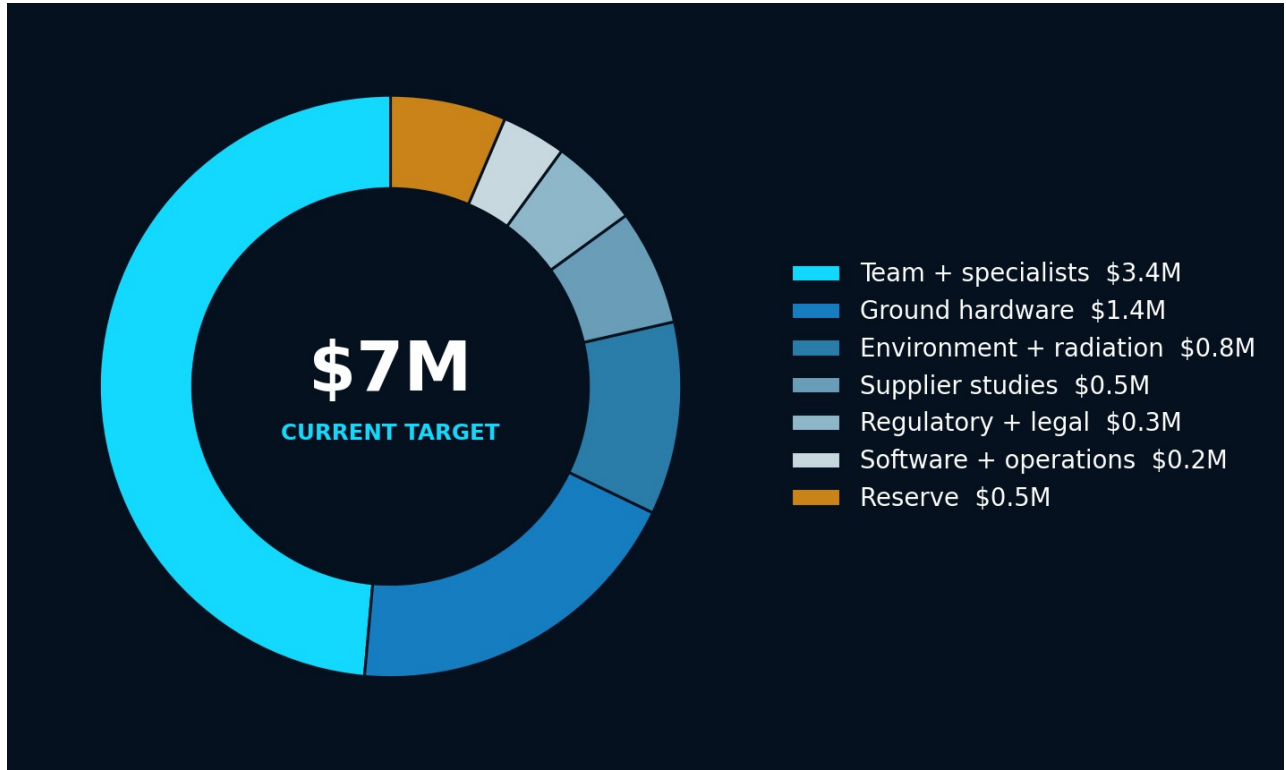
Gate	Required evidence	Decision
MDR	Mission need, three workloads, sensor/data route, 10 kW requirement rationale, initial orbit trade	Authorize SRR work
SRR	Requirements, 1 kW tile results, interface assumptions, supplier responses, preliminary licensing path	Freeze requirements baseline
PDR	10 kW breadboard, selected architecture, mass/power/thermal budgets, customer evidence and supplier ROMs	Authorize detailed design and flight-development raise
CDR	Released design, analyses, qualification plan, launch interface, software and security closure	Authorize flight build
FRR	Qualification and acceptance results, operations, licensing and launch readiness	Authorize launch

Commercial gates

- Before mission freeze: 20 structured buyer and data-owner interviews plus representative datasets for three workloads.
- Before SRR: two paid ground evaluations, one credible sensor-ingress partner and measured end-to-end benchmarks.
- Before PDR: one conditional anchor reservation representing at least 25% of base-case required annual revenue and two additional paid design partners.
- Before non-refundable launch commitment: binding reservations covering at least 50% of first-three-year target revenue and risk-adjusted contracted lifetime revenue at least 1.5 times deployed mission cost.

\$7 MILLION DEVELOPMENT ROUND

10. Current financing plan



Use	\$M	Deliverable
Core engineering team and specialists	3.40	Systems, power, thermal, compute, software and program leadership
Compute, power and thermal ground hardware	1.35	1 kW tile, 10 kW breadboard, facility cooling and instrumentation
Environmental and radiation testing	0.75	Qualification planning and characterization of critical parts
Supplier studies, interfaces and preliminary integration	0.45	ROMs, ICD inputs, analyses and gate reviews
Regulatory, export, security and legal	0.35	Early licensing and compliance path
Software, data, facilities and operations	0.25	Runtime, workloads, customer evaluations and company infrastructure
Program reserve	0.45	Schedule, supplier, test and hiring uncertainty
Total	7.0	Development evidence through a credible PDR target

Staged capital, not one speculative \$100 million ask

Stage	Planning range	What must be true first
Current development round	\$7M	Founding team, ground evidence, customer evidence, supplier ROMs, SRR and PDR
Flight-development round	\$25-40M	PDR-quality design, anchor evidence and credible integration path
First-node program financing	\$100-150M	Supplier-backed build and launch plan plus material contracted demand
Fleet expansion	\$200M+	Stable operations, paid utilization, replication economics and manufacturing plan

These are planning ranges rather than executable financing terms. The instrument mix may include venture equity, strategic capital, government non-dilutive funding, customer-funded engineering, milestone prepayments and later asset or project financing. Legal, tax and securities counsel must review any offering or public communication.

SELL OUTCOMES, NOT KILOWATTS

11. Commercial model and first-node economics

The recommended customer agreement is an Assured Orbital Processing Agreement. Kilowatts remain an internal capacity measure. Customer pricing should be expressed through reserved throughput, response time, mission availability, accepted outputs and rights to priority service.

1. Paid evaluation and integration order: benchmark representative data and define acceptance metrics.
2. Conditional capacity reservation: customer reserves future service subject to commissioning and performance.
3. Operational assured service: multi-year take-or-pay commitment with priority, minimum throughput and SLAs.
4. Usage and outcome charges: additional processing, crosslink, storage, egress and specialized derived products.

Illustrative revenue requirement

Scenario	Deployed cost	Annual operations	Life	Target GM	Required annual revenue
Low	\$60M	\$4M	5 years	40%	\$26.7M
Base	\$100M	\$6M	5 years	40%	\$43.3M
High	\$150M	\$10M	5 years	40%	\$66.7M

These are internal scenario assumptions, not quotes or a valuation. They show why terrestrial GPU-hour pricing is insufficient and why an anchor customer must pay for assured mission outcomes. The model must be replaced by supplier-backed P50 and P90 lifecycle costs before PDR.

ECONOMIC**IMPLICATION**

At 70% paid utilization, the illustrative cases require roughly \$3.8 million to \$9.5 million per sold kW-year. This is a warning against pricing the system as commodity compute, not a proposed tariff.

WHAT CAN BREAK THE PROGRAM

12. Key risks and design responses

Risk	Why it matters	Current response	Next closure evidence
Thermal rejection	10 kW compute becomes roughly 10 kW of heat	Conservative radiator screen and ground heat-load rig	Orbital thermal model and supplier-backed hardware
Data ingress	A free-flyer without sensor data is idle	Same-platform and crosslink alternatives	Partner interface and end-to-end demonstration
Market demand	No public 10 kW capacity procurement exists	Outcome-based customer discovery	Paid pilots and conditional reservation
Radiation and reliability	COTS performance and mission life are uncertain	Tile isolation, ECC, watchdogs and spares	Environment model and component testing
Deployment dynamics	Large arrays and radiators drive structure and control	25-40 m span planning envelope	CAD, modal and deployment analyses
Launch and integration	3-7 t cannot use small-rideshare list prices	Heavy rideshare or dedicated trade	Mass properties and launch-provider study
Capital timing	Flight program exceeds current seed	Gate-based rounds and reserve	PDR evidence and contracted demand
Export, security and licensing	Defense, sensors, crypto and spacecraft are regulated	Early counsel and data classification	Jurisdiction-specific compliance plan

PLAN BEFORE LAUNCH COMMITMENT

13. Regulatory and sustainability framework

The exact licensing package depends on spacecraft ownership, operator location, sensor control, frequency use, launch arrangement, data products and customer. The following workstreams are expected, but this document does not provide legal advice.

Workstream	Likely authority or owner	Trigger
Space-station and spectrum authorization	FCC or relevant national regulator	U.S.-licensed communications and spacecraft operations
Launch or reentry licensing	FAA AST and launch provider	U.S. launch or reentry activities; provider integrates customer requirements
Commercial remote sensing	NOAA CRSRA where applicable	Operation of a U.S. private remote-sensing system
Export controls	BIS, DDTC where applicable, counsel and suppliers	Technical data, hardware, software, foreign persons and destinations
Space-object registration and notifications	Licensing state and national authorities	Operator and launching-state responsibilities
Debris mitigation and disposal	Licensing authority plus program	Orbit selection, reliability, maneuverability, passivation and end-of-life
Customer data and security	Contracting authority, customer and accreditation owner	Defense, sovereign, controlled or sensitive workloads

The baseline should avoid intentional long-lived debris, include positive deorbit or rapid natural decay where feasible, provide conjunction-data support and preserve passivation after credible faults. Disposal capability is a flight requirement, not a public-relations add-on.

SOURCES AND LIMITS

14. Public evidence ledger

Evidence	What it supports	What it does not prove
NASA power and thermal surveys	Need to size EOL generation, eclipse storage, heat transport and radiators	This exact architecture or supplier performance
Apex Comet Mini public page	Commercial platforms approach high payload-power classes	10 kW continuous compute plus 10 kW heat rejection
EnduroSat FRAME Max public page	Multi-tonne payload and high-generation platform class exists	Turnkey continuous 10 kW payload service
NASA and ESA onboard-AI missions	Space-originated data can be processed before downlink	Demand for a 10 kW independent node
GAO and BlackSky disclosures	Government and EO buyers fund assured data and analytics outcomes	Transferable price for compute capacity
Recent sector financings	Investors have funded orbital-compute and satellite-infrastructure companies	Satellite Inference valuation or investor demand

Primary and issuer sources

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THE HONEST BOUNDARY OF V0.2

15. Explicit unknowns and next work

Unknown	Owner or evidence source	Target gate
Exact workload mix and sustained duty cycle	Paid customer evaluations	MDR / SRR
Compute hardware, accelerator count and radiation strategy	Tile benchmark and radiation campaign	SRR / PDR
Bus, arrays, batteries, thermal loop and radiator architecture	Supplier studies and integrated model	PDR
Final orbit, annual beta history and attitude profile	Mission analysis	SRR / PDR
Crosslink vendor, data rates, acquisition geometry and price	Vendor study plus partner interface	PDR
Mass properties, stowed volume and launch compatibility	CAD, structural analysis and launch-provider review	PDR / CDR
Licensing, export classification and security accreditation	Counsel, regulators, suppliers and customer	Before binding procurement
Supplier-backed P50/P90 program cost and schedule	ROMs, quotes and integrated master schedule	PDR
Price and binding customer demand	Paid pilots and capacity reservation	Before launch commitment

Conclusion

Satellite Inference should pursue a large first commercial target without pretending it is already designed. The program's differentiator is not simply putting a GPU in orbit. It is the integration of sensor access, continuous 10 kW computing, heat rejection, communications, fault-tolerant software and mission-assured customer delivery.

The next value-creating step is not a launch booking. It is a \$7 million evidence program that converts the 10 kW concept into a PDR-quality architecture, validated workloads, supplier-backed interfaces and customer commitments. Only then should the company underwrite the substantially larger capital required to build and launch the first node.